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NARAYANA
IIT-JEE ACADEMY-INDIA

SR STAR CO SC(MODEL-A&B)

Duration: 1 Hr

Date: 18.07.2024

Max. Marks: 200

CHEMISTRY NCERT TEST – 3

General Instructions:

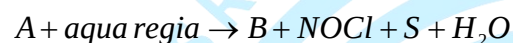
- **Attempt all the questions, each correct answer carries +4 and wrong answer -1.**

SYLLABUS: QUALITATIVE ANALYSIS (SALT ANALYSIS)

1. On reaction of Lead Sulphide with dilute nitric acid which of the following is not formed?

- 1) Lead nitrate 2) Sulphur 3) Nitric oxide 4) Nitrous oxide

2. Consider the following test for a group-IV cation.



The spin only magnetic moment value of the metal complex C is _____ BM. (Nearest integer)

- 1) zero 2) 1.732 BM 3) 2.89 BM 4) 4.98 BM

3. Match List I with List II

List I

(Name of the test)

(A) Borax bead test

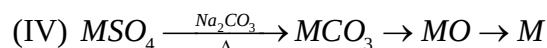
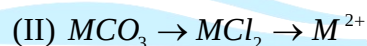
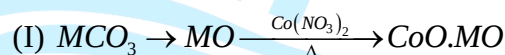
(B) Charcoal cavity test

(C) Cobalt nitrate test

(D) Flame test

List II

(Reaction sequence involved [M is metal])



Choose the correct answer from the options below:

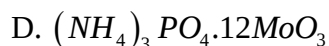
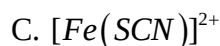
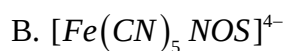
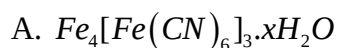
- 1) A(III), B(I), C(IV), D(II)
 2) A(III), B(II), C(IV), D(I)
 3) A(III), B(I), C(II), D(IV)
 4) A(III), B(IV), C(I), D(II)

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4. Match List-I with List-II:

List-I

(Compound)



List-II

(Colour)

(I) Violet

(II) Blood Red

(III) Prussian Blue

(IV) Yellow

Choose the correct answer from the options given below:

1) A-III, B-I, C-II, D-IV

2) A-IV, B-I, C-II, D-III

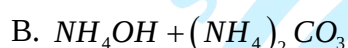
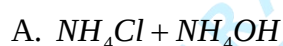
3) A-II, B-III, C-IV, D-I

4) A-I, B-II, C-III, D-IV

5. Match List I with List II

List I

(Precipitating reagent and conditions)



D. dilute HCl

List II

(Cation)

(I) Mn^{2+}

(II) Pb^{2+}

(III) Al^{3+}

(IV) Sr^{2+}

Choose the correct answer from the options given below:

1) A-IV, B-III, C-II, D-I

2) A-IV, B-III, C-I, D-II

3) A-III, B-IV, C-I, D-II

4) A-III, B-IV, C-II, D-I

6. During the detection of acidic radical present in a salt, a student gets a pale yellow precipitate soluble with difficulty in NH_4OH solution when sodium carbonate extract was first acidified with dil. HNO_3 and then $AgNO_3$ solution was added. This indicates presence of

1) Br^-

2) CO_3^{2-}

3) I^-

4) Cl^-

7. In the precipitation of the iron in group (III) in qualitative analysis, ammonium chloride is added before adding ammonium hydroxide to:

1) prevent interference by phosphate ions

2) decrease concentration of OH^- ions

3) increase concentration of Cl^- ions

4) increase concentration of NH_4^+ ions

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8. **Statement I:** Dimethyl glyoxime forms a six-membered covalent chelate when treated with $NiCl_2$ solution in presence of NH_4OH .
- Statement II:** Prussian blue precipitate contains iron in both (+2) and (+3) oxidation states.
- In the light of the above statements, choose the most appropriate answer from the options given below:
- 1) Statement I is false but statement II is true
 - 2) Both statement I and statement II are true
 - 3) Statement I is true but statement II is false
 - 4) Both statement I and statement II are false
9. The compound that is white in colour is
- 1) lead sulphide
 - 2) lead sulphate
 - 3) lead chromate
 - 4) lead iodide
10. **Statement I:** The gas liberated on warming a salt with dil. H_2SO_4 , turns a piece of paper dipped in lead acetate into black. It is a confirmatory test for sulphide ion.
- Statement II:** In statement I the colour of paper turns black because of formation of lead sulphite.
- In the light of the above statements, choose the most appropriate answer from the options given below:
- 1) Both the statement I and statement II are false
 - 2) Statement I is false but statement II is true
 - 3) Statement I is true but statement II is false
 - 4) Both the statement I and statement II are true
11. A and B formed in the following reactions are:
- $$CrO_2Cl_2 + 4NaOH \rightarrow A + 2NaCl + 2H_2O, A + 2HCl + 2H_2O_2 \rightarrow B + 3H_2O$$
- 1) $A = Na_2CrO_4, B = CrO_5$
 - 2) $A = Na_2Cr_2O_4, B = Cr_2O_3$
 - 3) $A = Na_2Cr_2O_7, B = CrO_3$
 - 4) $A = Na_2Cr_2O_7, B = CrO_5$
12. A reagent which gives brilliant red precipitate with Nickel ions in basic medium is
- 1) dimethylglyoxime
 - 2) meta-dinitrobenzene
 - 3) sodium nitroprusside
 - 4) neutral $FeCl_3$
13. On passing a gas, 'X', through Nessler's reagent, a brown precipitate is obtained. The gas 'X' is
- 1) Cl_2
 - 2) H_2S
 - 3) CO_2
 - 4) NH_3
14. $NO_2^- + CH_3COOH \rightarrow A + CH_3COO^-$
 $A + 2KI + 2CH_3COOH \rightarrow 2CH_3COOK + 2H_2O + 2B + C$
- Choose the incorrect statement about the above sequence of reactions.
- 1) 'A' is an acid used diazotization
 - 2) 'B' as an odd electron species with reddish brown colour
 - 3) 'C' forms blue complex with starch
 - 4) KI acts as a reducing agent

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15. In the wet tests for detection of various cations by precipitation, Ba^{2+} cations are detected by obtaining precipitation of
- $Ba(ox)$: Barium oxalate
 - $BaCO_3$
 - $Ba(OAc)_7$
 - $BaSO_4$
16. When a solution of mixture having two inorganic salts was treated with freshly prepared ferrous sulphate in acidic medium, a dark brown ring was formed whereas on treatment with neutral $FeCl_3$, it gave deep red colour which disappeared on boiling and a brown red ppt was formed. The mixture contains.
- CH_3COO^- & NO_3^-
 - $C_2O_4^{2-}$ & NO_3^-
 - SO_3^{2-} & CH_3COO^-
 - SO_3^{2-} & $C_2O_4^{2-}$
17. Formation of which complex, among the following is not a confirmatory test of Pb^{2+} ions
- lead chromate
 - lead iodide
 - lead nitrate
 - lead sulphate
18. Element not present in Nessler's reagent is
- Hg
 - I
 - K
 - N
19. During the qualitative analysis of SO_3^{2-} using dilute H_2SO_4 , SO_2 gas is evolved which turns $K_2Cr_2O_7$ solution (acidified with dilute H_2SO_4)
- Black
 - Red
 - Green
 - Blue
20. During the borax bead test with $CuSO_4$, a blue green colour of the bead was observed in oxidising flame due to the formation of
- Cu_3B_2
 - Cu
 - $Cu(BO_2)_2$
 - CuO
21. Match the List I with List II:
- | List I (Cations) | List II (Group reaction) |
|----------------------------------|---|
| $P \rightarrow Pb^{2+}, Cu^{2+}$ | (i) H_2S gas in presence of dilute HCl |
| $Q \rightarrow Al^{3+}, Fe^{3+}$ | (ii) $(NH_4)_2CO_3$ in presence of NH_4OH |
| $R \rightarrow Co^{2+}, Ni^{2+}$ | (iii) NH_4OH in presence of NH_4Cl |
| $S \rightarrow Ba^{2+}, Ca^{2+}$ | (iv) H_2S in presence of NH_4OH |
- $P \rightarrow i, Q \rightarrow iii, R \rightarrow ii, S \rightarrow iv$
 - $P \rightarrow iv, Q \rightarrow ii, R \rightarrow iii, S \rightarrow i$
 - $P \rightarrow iii, Q \rightarrow i, R \rightarrow iv, S \rightarrow ii$
 - $P \rightarrow i, Q \rightarrow iii, R \rightarrow iv, S \rightarrow ii$
22. Cation 'X' belongs to VI group. This cation gives a precipitated with disodium hydrogen phosphate solution after scratching the inner walls of the test tube. The cation is
- NH_4^+
 - Mg^{2+}
 - K^+
 - Na^+
23. A white precipitate was formed when $BaCl_2$ was added to water extract of an inorganic salt. Further, a gas 'X' with characteristic odour was released when the formed white precipitate was dissolved in dilute H_2SO_4 . The anion present in the inorganic salt is

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- 1) I^- 2) SO_3^{2-} 3) S^{2-} 4) NO_2^-

24. Match List I with List II

List I (Anion)

List II (Gas evolved on reaction with dil H_2SO_4)

- (A) CO_3^{2-} (i) Colourless gas which turns lead acetate paper black
 (B) S^{2-} (ii) Colourless gas which turns acidified potassium dichromate solution green.
 (C) SO_3^{2-} (iii) Brown fumes which turns acidified KI solution containing starch blue.
 (D) NO_2^- (iv) Colourless gas evolved with brisk effervescences, which turns lime water milky.

Choose the correct answer from the options given below:

- 1) A-III, B-I, C-II, D-IV 2) A-IV, B-I, C-IV, D-III
 3) A-IV, B-I, C-III, D-II 4) A-IV, B-I, C-II, D-III

25. In the flame test of a mixture of salts, a green flame with blue centre was observed. Which one of the following cations may be present?

- 1) Cu^{2+} 2) Sr^{2+} 3) Ba^{2+} 4) Ca^{2+}

26. Reagent, 1-naphthylamine and sulphanilic acid in acetic acid is used for the detection of

- 1) NO_3^- 2) N_2O 3) NO_2^- 4) NO

27. The incorrect statement out of the following is

- 1) Cu^{2+} ion gives chocolate coloured precipitate with potassium ferrocyanide solution
 2) Cu^{2+} and Ni^{2+} ions give black precipitate with H_2S in presence of HCl solution
 3) Ferric ion gives blood red opaque colour with potassium thiocyanate
 4) Cu^{2+} salts give red coloured borax bead test in reducing flame in cold condition

28. A solution containing a group-IV cation gives a precipitate on passing H_2S . A solution of this precipitate in dil. HCl produces a white precipitate with NaOH solution and bluish-white precipitate with basic potassium ferrocyanide. The cation is

- 1) Co^{2+} 2) Ni^{2+} 3) Mn^{2+} 4) Zn^{2+}

29. An aqueous solution of a salt X turns blood red on treatment with SCN^- and blue on treatment with $K_4[Fe(CN)_6]$. X also gives a positive chromyl chloride test. The salt X is

- 1) $CuCl_2$ 2) $FeCl_3$ 3) $Cu(NO_3)_2$ 4) $Fe(NO_3)_3$

30. A pink coloured salt turns blue on heating. The presence of which cation is most likely?

- 1) Co^{2+} 2) Cu^{2+} 3) Zn^{2+} 4) Fe^{2+}

31. Sodium carbonate cannot be used in place of $(NH_4)_2CO_3$ for the identification of

Ca^{2+} , Ba^{2+} and Sr^{2+} ions (in group V) during mixture analysis because.:

- 1) Mg^{2+} ions will also be precipitated
 2) Concentration of CO_3^{2-} ions is very low

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- 3) Sodium ions will react with acid radicals
- 4) Na^+ ions will interfere with the detection of Ca^{2+} , Ba^{2+} , Sr^{2+} ions.
32. Which of the following pairs of acid radicals is decomposed by dil. H_2SO_4 ?
- 1) NO_2^- , I^- 2) SO_3^{2-} , CO_3^{2-} 3) CO_3^{2-} , Cl^- 4) Br^- , I^-
33. A halide salt on heating with conc. H_2SO_4 and MnO_2 liberates reddish brown vapours of
- 1) Br_2 2) F_2 3) I_2 4) Cl_2
34. The brown colour ring during the test of NO_3^- is due to the formation of complex ion with the formula
- 1) $[Fe(H_2O)_6]^{2+}$ 2) $[Fe(NO)(CN)_3]^{2+}$
- 3) $[Fe(H_2O)_5(NO)]^{2+}$ 4) $[Fe(H_2O)(NO)_5]^{2+}$
35. A substance on treatment with dil. H_2SO_4 liberates a colourless gas which produces (i) Turbidity with baryta water and (ii) turns acidified dichromate solution green. The reaction indicates the presence of
- 1) $C_2O_3^{2-}$ 2) S^{2-} 3) SO_3^{2-} 4) NO_2^-
36. Experiment of detection of S^{2-} is given below in alkaline medium
- $Na_2S + [Fe(CN)_3NO]^{2-} \rightarrow X$ complex product
- Select the **correct** statement about the experiment
- 1) Colour of the complex compound formed in product is purple colour
- 2) Na^+ ions acts as a central metal ion in the complex
- 3) IUPAC name of complex compound formed in product is pentacyanonitroferrate (I) ion
- 4) It is a test to detect NO
37. Oxidation state of Molybdenum in ammonium phosphomolybdate is
- 1) +3 2) +6 3) +4 4) +2
38. The gas produced in the reaction given below at $25^\circ C$ is
- $Na_2C_2O_4 + H_2SO_4 \xrightarrow{(Conc.)}$
- 1) SO_2 only 2) CO_2 only 3) CO_2 and CO both 4) Na_2SO_4
39. Match the columns

Precipitate	Colour
(A) CuS	(p) reddish brown
(B) $Al(OH)_3$	(q) White
(C) $Fe(OH)_3$	(r) White gelatinous
(D) $Mn(OH)_3$	(s) black

- 1) A-s; B-r; C-p; D-q
3) A-p; B-q; C-r; D-s

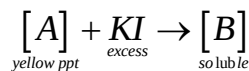
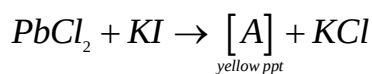
- 2) A-r; B-s; C-p; D-q
4) A-s; B-p; C-q; D-r

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40. A small amount of NH_4OH precipitates basic copper sulphate which is soluble in excess of NH_4OH due to the formation of

- 1) Tetraammine copper (I) sulphate
- 2) Tetraaquo copper (II) sulphate
- 3) Tetraammine copper (II) sulphate
- 4) Hexaammine copper (II) sulphate

41. Consider the following reactions :



Compound [A] and [B] are

- 1) PbI_4 and $K_2[PbI_4]$ respectively
- 2) $K_2[PbI_4]$ and PbI_4 respectively
- 3) PbI_2 and $K_2[PbI_4]$ respectively
- 4) PbI_2 and $K_2[PbI_2]$ respectively

42. Match the column – I with column-II and choose the correct option

	Column – I		Column - II
a.	Group- VI	i)	Fe^{3+}
b.	Group- V	ii)	Mg^{2+}
c.	Group- I	iii)	Sr^{2+}
d.	Group- III	iv)	Pb^{2+}

- 1) a(ii), b(iii), c(iv), d(i)
- 2) a(i), b(iii), c(iv), d(ii)
- 3) a(iv), b(iii), c(ii), d(i)
- 4) a(ii), b(i), c(iv), d(iii)

43. $Fe(OH)_3$ can be separated from $Al(OH)_3$ by addition of

- 1) Dil. HCl
- 2) NaCl solution
- 3) NaOH solution
- 4) NH_4Cl and NH_4OH

44. An aqueous solution containing a mixture of copper (II), iron (II) and lead (II) ions was treated with an excess of aqueous ammonia. Which precipitate was left after adding of excess of NH_3 ?

- 1) Copper (II) hydroxide only
- 2) Iron (II) hydroxide only
- 3) Lead (II) hydroxide only
- 4) Lead (II) hydroxide and iron (II) hydroxide

45. A black sulphide is formed by the action of H_2S on

- 1) $CuCl_2$
- 2) $CdCl_2$
- 3) $ZnCl_2$
- 4) NaCl

46. Few drops of HNO_3 are added to II group before proceeding to III group in order to

- 1) Convert Fe^{+2} to Fe^{+3}
- 2) Convert Fe^{+3} to Fe^{+2}
- 3) ppt III group
- 4) convert Fe^{+2} to Fe

47. Sulphuric acid is not used for preparation of original solution in analysis of basic radicals because

- 1) it is a reducing agent
- 2) it forms insoluble sulphate with certain basic radicals
- 3) it forms a soluble complex
- 4) it is viscous in nature

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48. The reddish brown ppt formed in the Nessler's test for NH_4^+

- 1) $NH_2HgOHgI$ 2) NH_2HgOHI_2 3) $NH_3HgOHgI_2$ 4) $NH_3HgOHgI$

49. Match the following

	Column – I (salt)		Column – II (colour of the flame by naked eye)
1)	Strontium	a)	Brick red
2)	Barium	b)	Apple green
3)	Calcium	c)	Crimson red
4)	Magnesium	d)	No flame colour

1) a-(1), b-(2), c-(3), d-(4)

2) a-(3), b-(2), c-(1), d-(4)

3) a-(4), b-(3), c-(1), d-(2)

4) a-(1), b-(3), c-(2), d-(4)

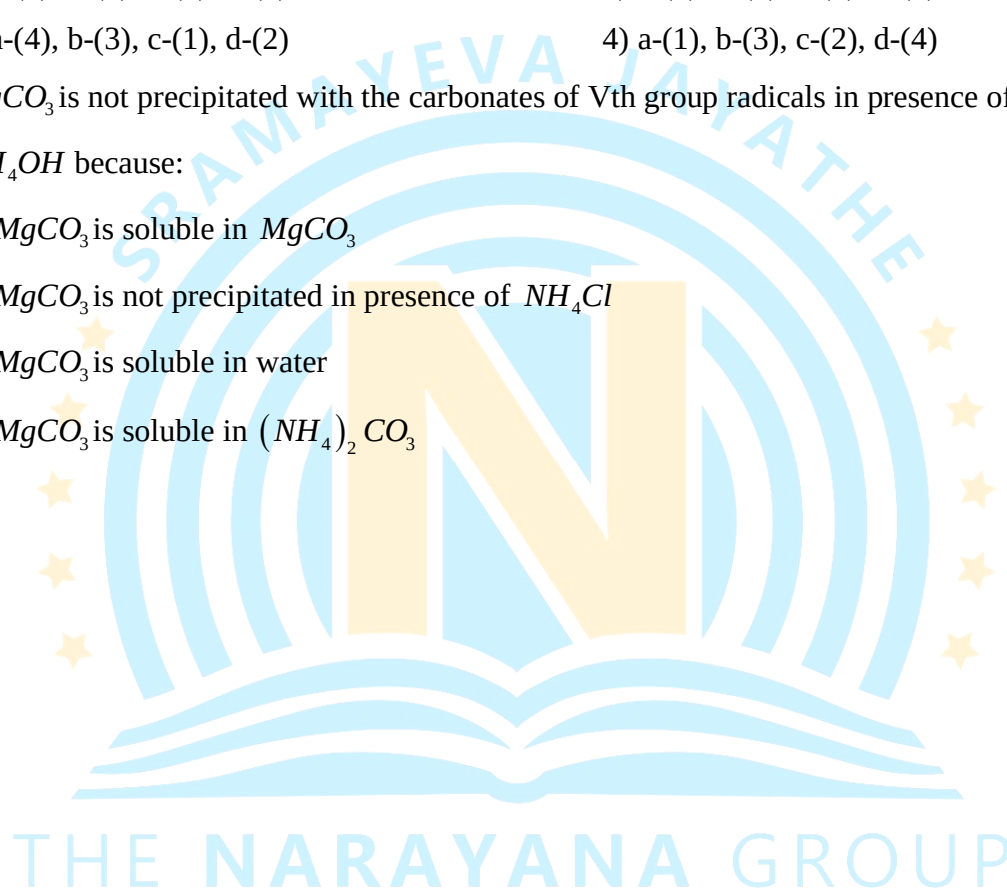
50. $MgCO_3$ is not precipitated with the carbonates of Vth group radicals in presence of NH_4Cl and NH_4OH because:

1) $MgCO_3$ is soluble in $MgCO_3$

2) $MgCO_3$ is not precipitated in presence of NH_4Cl

3) $MgCO_3$ is soluble in water

4) $MgCO_3$ is soluble in $(NH_4)_2CO_3$



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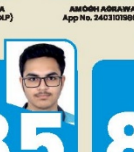
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